

PINIT-DNA — Test Cases Report

Scope: Pixel-level protection for images and video, and the Investigation/composition pipeline.

Compiled: 2026-09-15

Status legend: PASS = Pass / Fixed & verified · DIAGNOSED = root cause found, no code defect · FIXED = real bug found & fixed · OPEN = real bug found, still open · NOT RUN = not yet run

Part A — Automated Test Cases (Jest, npm test)

592 automated test cases exist in the repository’s Jest suite. **365** of them are specific to pixel-level image/video protection and the investigation-composition pipeline; the remainder cover the broader investigation/matching system (not pixel-level mechanics specifically, and not tied to a particular file type).

A1. Spatial Pixel-Auth Engine (HKCA) — 268 tests

The core pixel-level protection engine. Shared by images and every protected video frame alike (each frame runs through the same engine).

File	Tests	Covers
spatial-auth-phase4d-1x1.test.ts	45	Dense 1×1 pixel-level authentication (finest tier)
spatial-auth-phase3e-security.test.ts	39	Security hardening — tag forgery, replay, tamper-package attacks
spatial-auth-phase4e-hardening.test.ts	23	Hardening at the 1×1 hierarchy tier
spatial-auth-phase4c-2x2.test.ts	23	2×2 block hierarchy tier
spatial-auth-phase4b-4x4.test.ts	20	4×4 block hierarchy tier
spatial-auth-phase4a-hierarchy.test.ts	18	The 64→8→4→2→1 hierarchy structure itself
spatial-auth-phase3f-hardening.test.ts	17	HKCA (8×8 cell) hardening
spatial-auth-phase3c.test.ts	17	HKCA tamper-region clustering
spatial-auth-phase3b-validation.test.ts	17	HKCA validation edge cases
spatial-auth-phase2.test.ts	13	Phase 2 block-level auth
spatial-auth-phase1.test.ts	13	Phase 1 foundational block auth
spatial-auth-e2e-validation.test.ts	12	End-to-end validation across phases
spatial-auth-phase3a.test.ts	11	HKCA (8×8) core enrollment/verify

Representative individual cases (real test names): “One-pixel modification”, “Cross-image block transplant”, “Resize → DIMENSION_MISMATCH”, “JPEG recompression”, “Metadata-only EXIF change”, “Tampered SpatialAuthPackage / Merkle root → INVALID_AUTH_PACKAGE”, “position binding: same pixels different (x,y) produce different tags”, “boundary pixel → correct block(s) fail”, “full-image modification → GLOBAL pattern”, “mask bytes are block-accurate (no blur)”.

A2. The 6 DNA Layers — 60 tests

Run on every protected file, image or video frame alike.

File	Tests
layer5.metadata.test.ts	11
layer4.semantic.test.ts	11

layer6.steganography.test.ts	10
layer3.perceptual.test.ts	9
layer2.structural.test.ts	9
layer1.cryptographic.test.ts	7
ownership-watermark.test.ts	3

A3. Image & Video Composition / Pixel-Source-Map — 37 tests (full list)

The “how much is vault content, how much is your own” investigation logic.

Image composition — tests/forensics/investigation-composition.test.ts

1. splits probe so protected pixels are never counted as AI
2. puts unmatched remainder in Other when AI model is unavailable
3. uses fragment bbox coverage for a crop pasted into an AI image
4. falls back to local-DNA match/coverage ratios for crop-scale patch hits
5. keeps unmatched collage pixels GREY unless a separate AI detector is used
6. prefers a localized crop bbox over a collapsed 0.3% fragment island
7. ignores a collapsed fragment bbox so it is not shown as protected coverage
8. uses crop bbox when DNA fragments are missing but the paste is localized
9. does not treat a 56% whole-image lookalike as protected-file coverage
10. computes region area from width × height percents
11. exposes the three colored labels the investigation UI renders
12. uses block-level vault overlay percents when present
13. explains Vault origin without claiming a Vault ID lives in the pixel

Video composition — tests/forensics/video-investigation-composition.test.ts (added this session)

14. tiles the full probe duration with no gaps or overlap
15. collapses consecutive same-source frames into a single run
16. keeps a source change as separate timeline segments
17. weights overall percentages by segment duration, not a flat per-frame average
18. reports 0% protected / 100% other when nothing matched
19. averages originalUsedPercent only across frames that report it
20. handles a single frame spanning the whole duration

Pixel-source map — tests/forensics/pixel-source-map.test.ts

21. TEST 1 — exact original is ~100% vault pixels
22. TEST 2 — unrelated canvas has 0% vault pixels
23. TEST 3 — vault crop pasted into AI host is green only in the paste
24. TEST 4 — same crop at a new location is green at the new coordinates
25. TEST 7 — one pixel change is not green; the rest stay vault
26. TEST 8 — ten modified pixels inside a paste are not green

Forensic result states — tests/forensics/forensic-result-states.test.ts (added this session)

27. exposes five explicit states
28. MATRIX exact original + HMAC → VERIFIED_VAULT_ORIGIN / GREEN
29. MATRIX crop/copy-paste with spatial but no HMAC → LIKELY_VAULT_ORIGIN / GREEN
30. MATRIX JPEG/photometric: spatial holds, HMAC optional → still GREEN internally LIKELY
31. MATRIX mapped mismatch → NON_VAULT_MODIFIED / ORANGE, not AI
32. MATRIX unrelated camera/stock photo → UNKNOWN / GREY, not AI_SUSPECTED
33. MATRIX independent AI detector only → AI_SUSPECTED / ORANGE
34. MATRIX isolated / weak matches → UNKNOWN (do not fabricate certainty)

- 35. non-Vault is not a synonym for AI-generated
- 36. AI detector score is not used as Vault coverage
- 37. stamps region evidence for the expandable proof view

A4. Broader Investigation/Matching Pipeline — 128 tests

Across the remaining 16 tests/forensics/*.test.ts files: acceptance engine, candidate ranking, enterprise recovery pipeline, evidence pairing, DNA comparison, duplicate detection. Relevant to overall investigation correctness, not pixel-level mechanics specifically, and not image/video-specific.

Part B — Manual / User Test Cases (real browser, real clicks)

B1. Images

#	Test	Status	Result
1	Tamper localization heatmap (edit a region, re-investigate)	DIAGNOSED	The heatmap panel shown was a separate, older mechanism, not the new HKCA engine
2	Fragment/splice detection (eyes_spliced.jpg — a crop of the protected original pasted into unrelated content)	FIXED & VERIFIED	Real 80% confidence match found in the backend log; added a prominent top-of-report banner and fixed contrast bugs on evidence cards so it's actually visible
3	Protect-flow “stuck spinner” (upload → wait for success state)	DIAGNOSED	Backend was finishing DNA generation in under a minute; no live reproducible frontend bug found on final code re-read
4	False positive: unrelated landscape photo flagged as fragment reuse	FIXED & VERIFIED	Root cause was minPatchMatches defaulting to 3 (3 coincidental patches scored 68% confidence); raised to 6 in src/config/local-dna.ts ; true positive (eyes_spliced.jpg , 14–15 real patches) confirmed still passing after the fix

Fixtures used: scratch/tamper-test-images/ (base.jpg , cropped.jpg , resized.jpg , region_added.jpg , region_deleted.jpg , eyes_spliced.jpg) — all derived from one original so results are directly comparable.
Written report from this pass: docs/Pixel_Protection_Test_Report.pdf .

B2. Video (real-backend scripted tests — same rigor as manual, run directly against the real dev DB/ffmpeg since a browser round-trip per case would take minutes to hours)

#	Test	Status	Result
5	Every-frame protection on a synthetic 12-frame video	PASS	12/12 frames protected, each with its own DnaRecord + SpatialAuthPackage row, correct sequential timestamps
6	Background protection job doesn't block the upload response	PASS	store() returned in ~1–5s while frame protection continued for minutes after
7	Direct protection call vs. real production trigger path (A/B)	PASS	Both produced correct results; 210.6s vs. 134.3s — no meaningful regression from the real trigger path
8	Whole-frame temporal reuse (trim + re-encode a protected video)	PASS	15/16 sampled frames matched; sensible composition % and timeline
9	True negative (unrelated synthetic video)	PASS	Correctly returned no match
10	Real-world spatial crop (user's actual uploaded)	FIXED & VERIFIED	3 real bugs found & fixed: (1) video composition wasn't wired into most investigation code paths, (2) crop-candidate discovery searched the wrong pool of content, (3) a “too complete to

	video, cropped to left half)		be a real fragment” filter incorrectly zeroed out a fully-correct match. Final verified result: 100% protected, 34% of original used, correct source video identified.
11	Discovery-path performance (after correctness fix)	PASS (PERF OPEN)	Correctness maintained throughout optimization; latency reduced from ~16 min → ~13.6 min via result-reuse + bounded concurrency — still slow for a live UI request; flagged as an open follow-up, not resolved

Not Run This Session (Tooling Exists, Honest Gap)

- `scripts/test-tamper-detection.cjs` — a real-HTTP tamper-detection test script; only inspected (to understand fixture generation), never executed end-to-end.
- `scripts/sentinel-benchmark/` — a ~34-case detection-accuracy benchmark suite; never run, its output folder does not exist on disk.
- **No screen-recorded proof videos exist** for any of the above — no screen-recording capability was available; only console/log output, database queries, and code changes.

Summary Counts

Category	Count
Automated Jest tests — total	592
Automated Jest tests — pixel-level image/video specific	365
Manual/scripted real-system test cases (images + video)	10
Real bugs found & fixed this session	5
Real bugs found, still open	1